



## **Software Requirements Specification (SRS)**

**Course:** CSE 3411 – SAD Lab

**Project Title:** Nirapod Nibash Bangladesh

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# 1. Introduction

## Purpose

The purpose of this SRS is to define all functional and non-functional requirements for *Nirapod Nibash Bangladesh (NNB)* — a digital disaster shelter management and coordination system. It serves as a shared technical blueprint for developers, clients, and stakeholders to ensure effective disaster response in Bangladesh.

## Intended Audience

This document is intended for developers, system analysts, government authorities, NGOs, volunteers, and academic evaluators who will use it to understand the technical structure, requirements, and operational flow of NNB.

## Scope of the System

NNB is a web and mobile-based system that enables citizens to find nearby disaster shelters, allows volunteers to manage tasks, and helps government/NGO officials monitor shelter occupancy, manage resources, and issue emergency alerts. It integrates GIS mapping, SMS communication, and cloud databases for real-time coordination.

Boundaries: Covers shelter management, volunteer coordination, resource tracking, and alert systems — excluding physical logistics and weather forecasting.

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## 2. System Overview

### Product Perspective

NNB operates as an integrated cloud-based platform within the national disaster management ecosystem. It connects citizens, volunteers, and government agencies through web, mobile, and SMS interfaces.

### Product Functions

1. Shelter Information Management
2. Citizen Access Interface
3. Resource & Aid Management
4. Volunteer & Staff Coordination
5. Communication & Alert System
6. Offline & Synchronization Module
7. Analytics & Reporting

### User Classes and Characteristics

- **Citizens:** Low-tech users needing simple, Bangla-based interfaces and SMS fallback.
- **Volunteers:** Medium-skill users who handle shelter updates and field coordination.
- **Government Officials:** Advanced users managing dashboards and analytics.
- **NGOs:** Medium-skill users maintaining resource and aid data.
- **Emergency Call Centers:** Operators handling real-time distress messages.

### Operating Environment

- Cloud-based servers (AWS/Azure/GCP).
- Database: PostgreSQL/MySQL.

- Web: React.js; Mobile: Flutter.
- Devices: Android ≥8.0, iOS ≥12, modern browsers.

### **Design and Implementation Constraints**

- Must comply with Bangladesh ICT Act (2006) and GDPR.
- Support for Bangla and English.
- Lightweight, optimized app for low-end devices.
- Open-source technology preferred.

### **Assumptions and Dependencies**

- Reliable government shelter data.
- Cooperation among telecoms and NGOs.
- Stable data center infrastructure.

### **SDLC Approach**

Adopted the **Waterfall Model**, progressing through:

1. Requirement Analysis
  2. System Design
  3. Implementation
  4. Testing
  5. Deployment
  6. Maintenance
-

### 3. Requirements Elicitation & Analysis

#### Benchmark Analysis

Compared with Red Cross Shelter Finder, Sahana Eden, and ICRC RedSafe.

|                                              | <br>Nirapod Nibash<br>Bangladesh | <br>Sahana Eden | <br>Red Cross<br>Emergency App | <br>REDSAFE<br>(ICRC) |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Real-time Shelter Availability               | ✓                                                                                                                 | ✓                                                                                                 | ✓                                                                                                                 | ✗                                                                                                        |
| Disaster-specific Alerts                     | ✓                                                                                                                 | ✓                                                                                                 | ✓                                                                                                                 | ✗                                                                                                        |
| Shelter Mapping & Navigation                 | ✓                                                                                                                 | ✓                                                                                                 | ✓                                                                                                                 | ✓                                                                                                        |
| Secure Document Storage                      | ✓                                                                                                                 | ✗                                                                                                 | ✗                                                                                                                 | ✗                                                                                                        |
| Volunteer & Staff Management                 | ✓                                                                                                                 | ✗                                                                                                 | ✗                                                                                                                 | ✗                                                                                                        |
| Multilingual Support (incl. Bangla)          | ✓                                                                                                                 | ✗                                                                                                 | ✗                                                                                                                 | ✗                                                                                                        |
| Low-bandwidth<br>Offline Functionaality      | ✓                                                                                                                 | ✓                                                                                                 | ✓                                                                                                                 | ✓                                                                                                        |
| Advanced Analytics & Reporting               | ✓                                                                                                                 | ✓                                                                                                 | ✗                                                                                                                 | ✗                                                                                                        |
| Disaster Simulation &<br>Preparedness Drills | ✓                                                                                                                 | ✓                                                                                                 | ✗                                                                                                                 | ✗                                                                                                        |

#### Requirements Analysis

Goal: Centralized and automated disaster shelter management.

Scope: Multi-user real-time coordination platform.

#### Information Collected

Sources include government disaster reports, NGO field data, and citizen surveys.

#### Sources of Information

Documents, field observation, interviews, online research, and stakeholder discussions.

## Gap Analysis

Existing systems are manual and uncoordinated → NNB provides digital, real-time data integration.

| Criteria                    | Existing Solutions                  | Identified Gaps                          | Nirapod Nibash Approach                           |
|-----------------------------|-------------------------------------|------------------------------------------|---------------------------------------------------|
| 🏠 Shelter Availability Data | Data exists but outdated/incomplete | No real-time, verified updates           | ✅ Real-time, cloud-synced verified shelter status |
| 🌐 Local Language Support    | Mostly English, partial translation | No full Bangla or dialect coverage       | ✅ Full Bangla + regional dialect support          |
| 📶 Offline Access            | Requires internet                   | No offline maps or alerts during outages | ✅ Offline maps + SMS emergency alerts             |
| 👤 Volunteer Management      | Fragmented & manual                 | No integrated recruitment/tracking       | ✅ In-app volunteer onboarding & task tracking     |
| 🔒 Data Security             | Limited encryption                  | Location & personal data vulnerable      | ✅ End-to-end encrypted personal & location data   |
| 🛠 Preparedness Tools        | Only during crises                  | No pre-disaster drills/training          | ✅ Interactive drills + built-in training modules  |

## Feature List Fixation

1. Shelter Search & Registration
2. Volunteer Management
3. Resource Inventory Tracking
4. Alert & Communication System
5. Offline Operation
6. Analytics Dashboard

## 4. Functional Requirements

### FR-1: User Authentication and Authorization

Secure signup with OTP verification and multi-role access (Citizen, Volunteer, NGO, Admin).

### FR-2: Shelter Registration and Management

Add, update, and classify shelters (Safe / Partial / Full / Closed).

### FR-3: Shelter Search and Navigation

Citizens find nearest shelters via GPS or SMS.

### FR-4: Real-Time Status Update

Shelter occupancy auto-updated every 10 seconds.

### FR-5: Resource Inventory Management

Tracks food, medicine, clothing, and water distribution.



**FR-6: Volunteer Management**

Registration, task assignment, and performance tracking.

**FR-7: Emergency Communication and Alert System**

Two-way alerts and SOS messages via SMS/app.

**FR-8: Offline Operation and Data Sync**

Caches 50 records; syncs automatically once online.

**FR-9: Reporting and Analytics**

Generates visual reports and exports (PDF, CSV).

**FR-10: System Administration**

Dashboard for user control, monitoring, and backups.

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**5. External & Non-Functional Requirements****User Interfaces**

- Web and mobile apps with responsive design.
- Color-coded statuses:  Safe,  Partial,  Full.
- Accessible layout with Bangla/English support.

**Software Interfaces**

- RESTful APIs, PostgreSQL DBMS, Google Maps API, Government Registry API.

**Performance Requirements**

- Response  $\leq 5$  seconds.
- Supports 10,000 concurrent users.
- Availability  $\geq 99.5\%$ .

**Security Requirements**

- AES-256 encryption, RBAC, MFA.

- HTTPS & OAuth 2.0 protocols.

**Maintainability**

- Modular code, version control (GitHub).
  - Clear API and documentation for scalability.
- 

**6. Feasibility & Project Management Analysis**

**Feasibility Analysis**

**Technical:** Achievable via open-source tools (React, Flutter).

**Economic:** Low-cost cloud hosting feasible.

**Operational:** Supports government and NGO workflows.

**Human Resource:** Student developers with tech expertise.

**Schedule:** Structured per Waterfall SDLC.

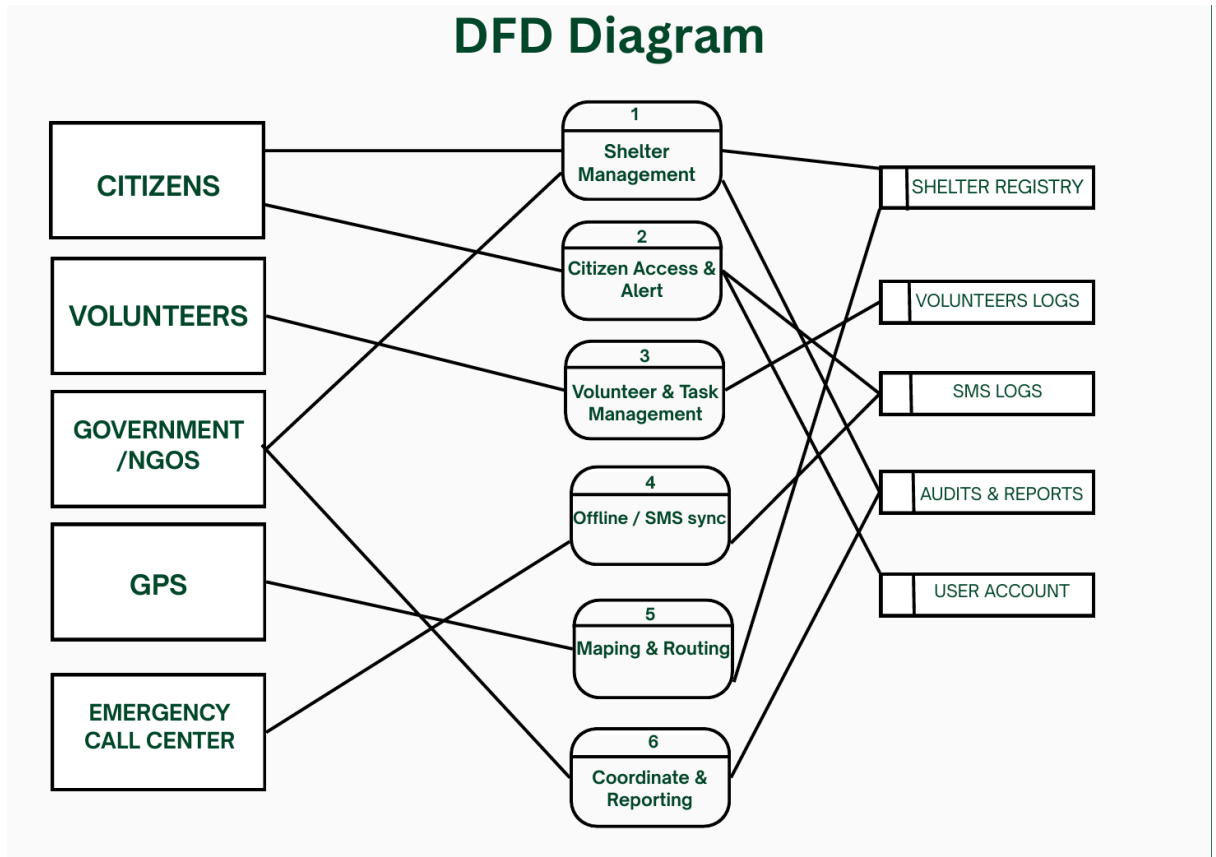
**SWOT Analysis**

| Strengths                      | Weaknesses            | Opportunities                 | Threats                 |
|--------------------------------|-----------------------|-------------------------------|-------------------------|
| Real-time centralized platform | Requires internet/SMS | National deployment potential | Power/network failure   |
| Multilingual & offline access  | Limited budget        | Integration with weather API  | Data-sharing reluctance |

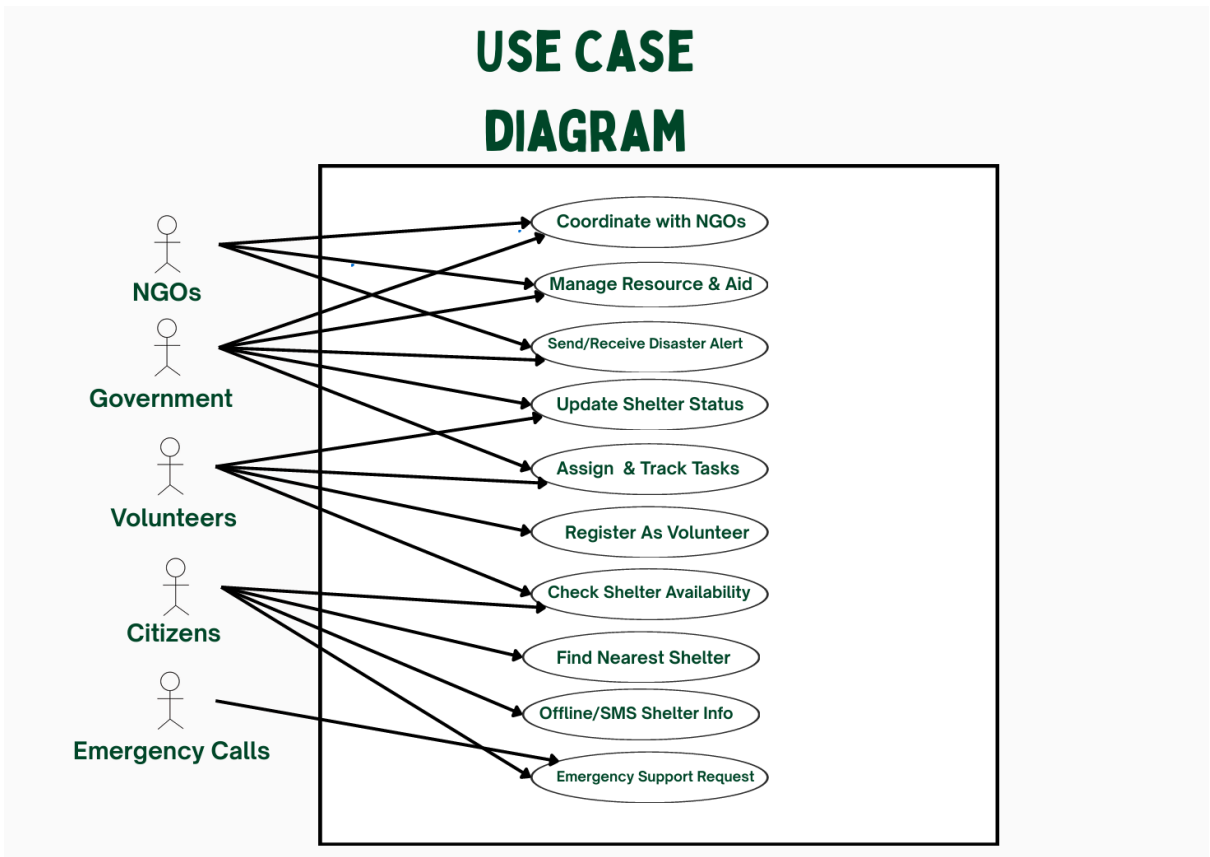
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## 7. System Design (Diagrams)

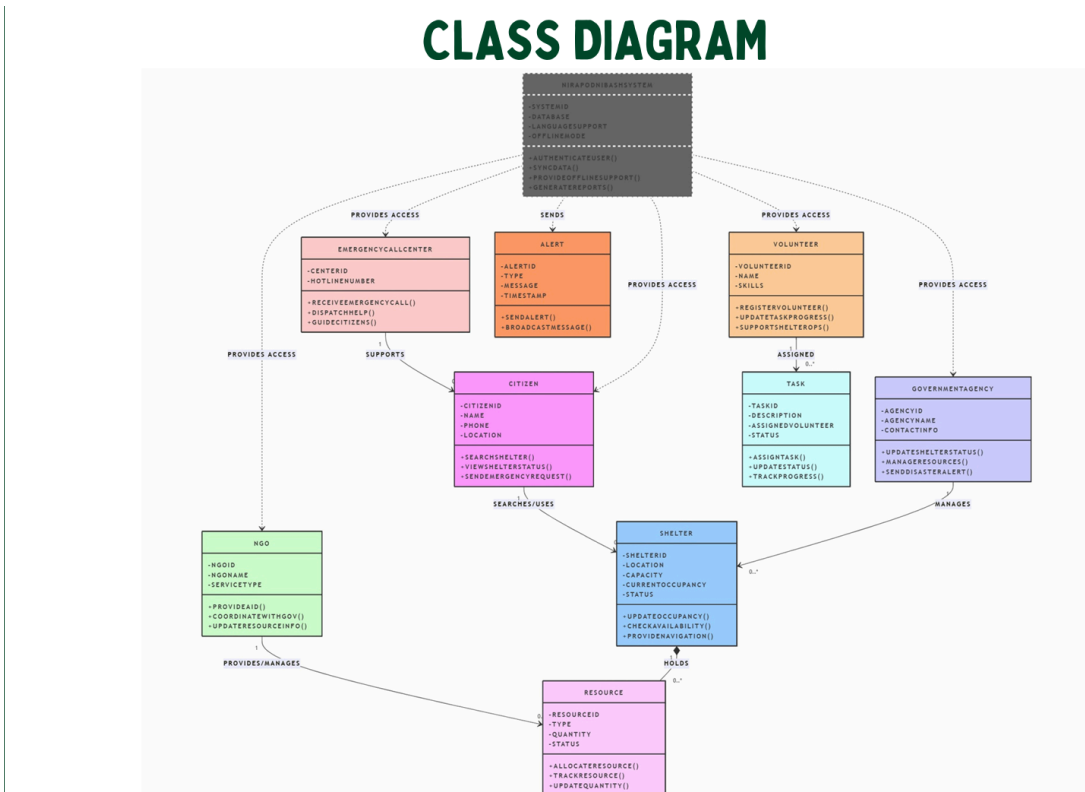
- Data Flow Diagram (DFD):



- Use Case Diagram:



- Class Diagram:



# SEQUENCE DIAGRAM

```

sequenceDiagram
    participant Citizen
    participant MobileApp
    participant SMSGateway
    participant GovernmentAgency
    participant NGO
    participant Volunteer
    participant EmergencyCallCenter
    participant Nirapod@bashSystem
    participant Database
    participant GPSService

    Citizen->>MobileApp
    Citizen->>SMSGateway
    Citizen->>GovernmentAgency
    Citizen->>NGO
    Citizen->>Volunteer
    Citizen->>EmergencyCallCenter
    Citizen->>Nirapod@bashSystem
    Citizen->>Database
    Citizen->>GPSService

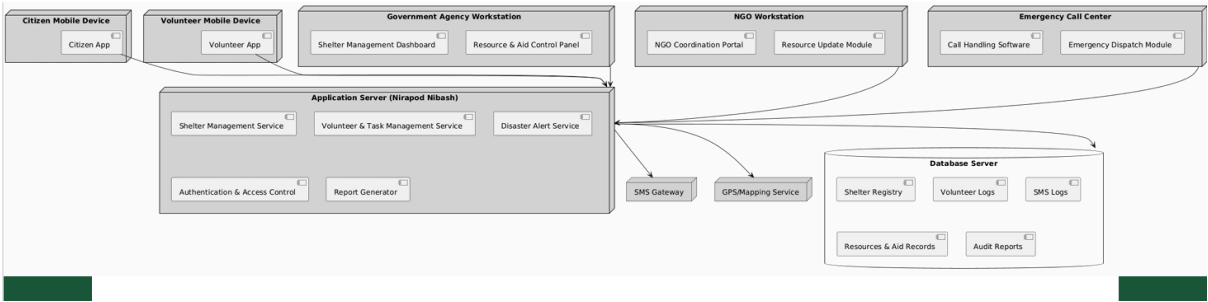
    GovernmentAgency->>Nirapod@bashSystem: SendDisasterAlert
    Nirapod@bashSystem->>MobileApp: BroadcastAlertMessage
    Nirapod@bashSystem->>SMSGateway: BroadcastAlertMessage
    Nirapod@bashSystem->>GovernmentAgency: RequestNearestShelterInfo
    Nirapod@bashSystem->>Database: QueryShelterAvailability
    Database->>Nirapod@bashSystem: ReturnShelterData
    Nirapod@bashSystem->>GovernmentAgency: ShowAvailableSheltersAndNavigation
    GovernmentAgency->>Nirapod@bashSystem: SendEmergencyHelpRequest
    Nirapod@bashSystem->>EmergencyCallCenter: LogEmergencyCase
    EmergencyCallCenter->>Volunteer: DispatchTaskAssignment
    Volunteer->>Nirapod@bashSystem: RegisterOrUpdateTaskProgress
    Nirapod@bashSystem->>Database: UpdateVolunteerLogs
    Volunteer->>Nirapod@bashSystem: UpdateResourceInformation
    Nirapod@bashSystem->>Database: StoreResourceInventory
    Nirapod@bashSystem->>GovernmentAgency: NotifyResourceUpdate
    GovernmentAgency->>Nirapod@bashSystem: AllocateResourcesToShelter
    Nirapod@bashSystem->>Database: UpdateShelterResourceStatus
    Nirapod@bashSystem->>GovernmentAgency: UpdateShelterInfo
    GovernmentAgency->>Nirapod@bashSystem: GenerateReportsAndSyncData
    Nirapod@bashSystem->>GovernmentAgency: GenerateReportsAndSyncData
    GovernmentAgency->>NGO: GenerateReportsAndSyncData
    NGO->>Volunteer: GenerateReportsAndSyncData
    Volunteer->>EmergencyCallCenter: GenerateReportsAndSyncData
    EmergencyCallCenter->>Nirapod@bashSystem: GenerateReportsAndSyncData
    Nirapod@bashSystem->>Database: StoreAuditLogsAndReports
  
```

# STATE DIAGRAM

```

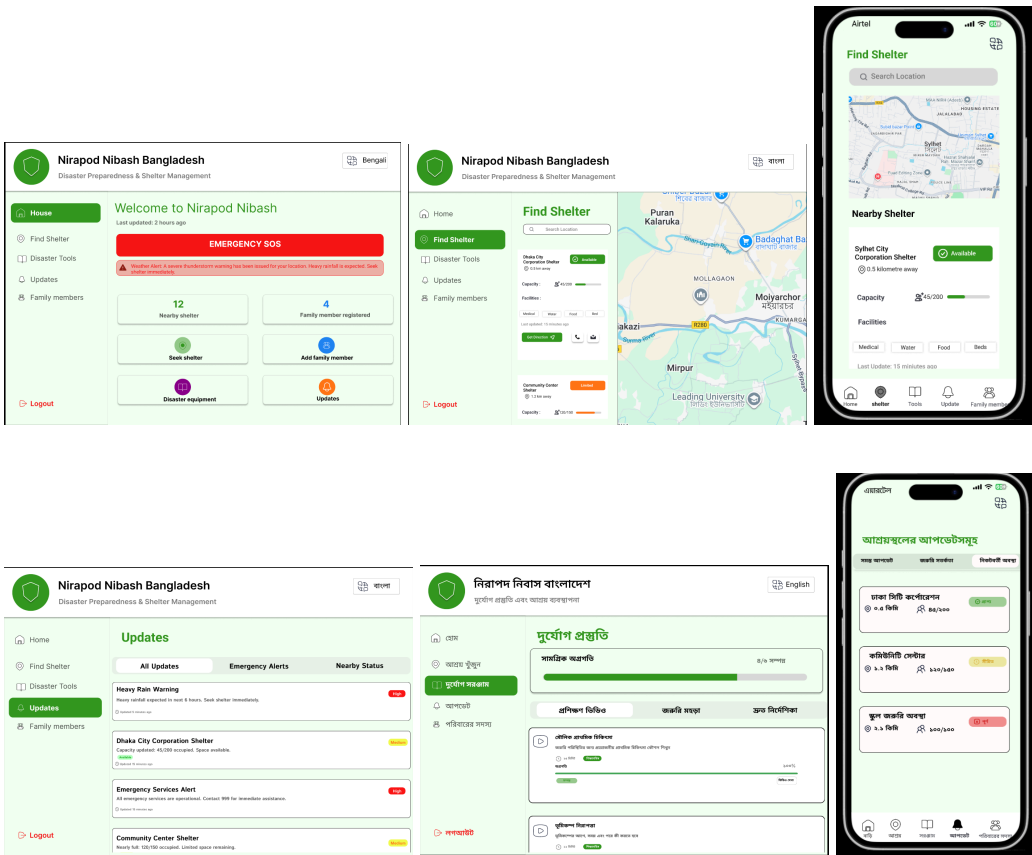
stateDiagram-v2
    [*] --> IDLENORMALSTATE
    IDLENORMALSTATE --> DISASTERALERTSTATE : DISASTER ALERT RECEIVED
    DISASTERALERTSTATE --> CITIZENALERTEDSTATE : NOTIFY CITIZENS
    CITIZENALERTEDSTATE --> SHELTERINFODISPLAYED : SEARCH SHELTER
    CITIZENALERTEDSTATE --> EMERGENCYCYSUPPORTSTATE : SEND EMERGENCY REQUEST
    CITIZENALERTEDSTATE --> IDLENORMALSTATE : END OF DISASTER
    SHELTERINFODISPLAYED --> CITIZENENROUTE : CITIZEN CHOOSES SHELTER
    CITIZENENROUTE --> CITIZENATSHELTER : ARRIVAL
    CITIZENATSHELTER --> SHELTERINFODISPLAYED : SHELTER FULL
    CITIZENATSHELTER --> SAFESTATE : DISPATCH HELP
    EMERGENCYCYSUPPORTSTATE --> SAFESTATE : DISPATCH HELP
    SAFESTATE --> IDLENORMALSTATE
    IDLENORMALSTATE --> VOLUNTEERTASKASSIGNED
    IDLENORMALSTATE --> RESOURCEMANAGEMENTSTATE
    VOLUNTEERTASKASSIGNED --> TASKCOMPLETEDSTATE : UPDATES PROGRESS
    TASKCOMPLETEDSTATE --> [*]
    RESOURCEMANAGEMENTSTATE --> RESSOURCEALLOCATION : RESOURCES LOW
    RESSOURCEALLOCATION --> [*]
  
```

● **Deployment Diagram:**

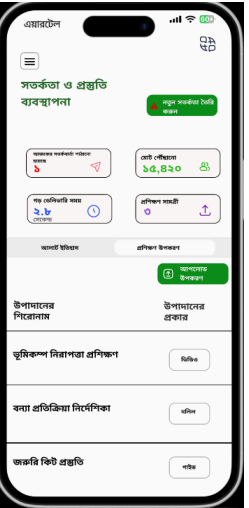
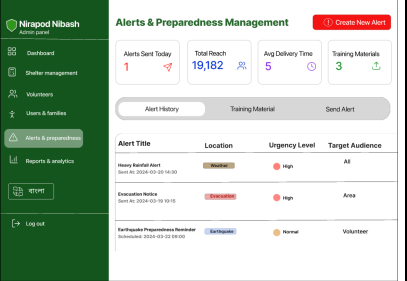
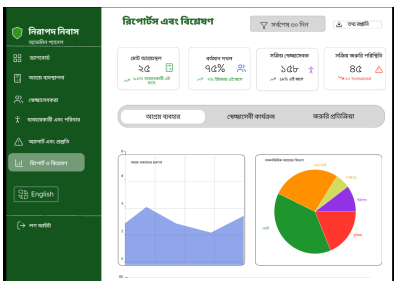
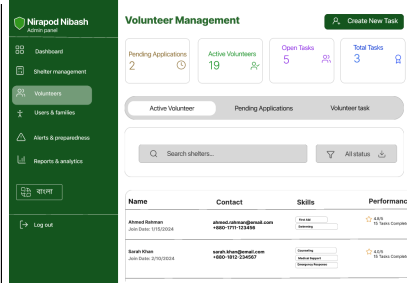
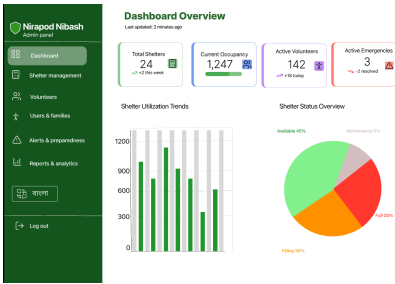


# 8. Prototype Design

- User Interface

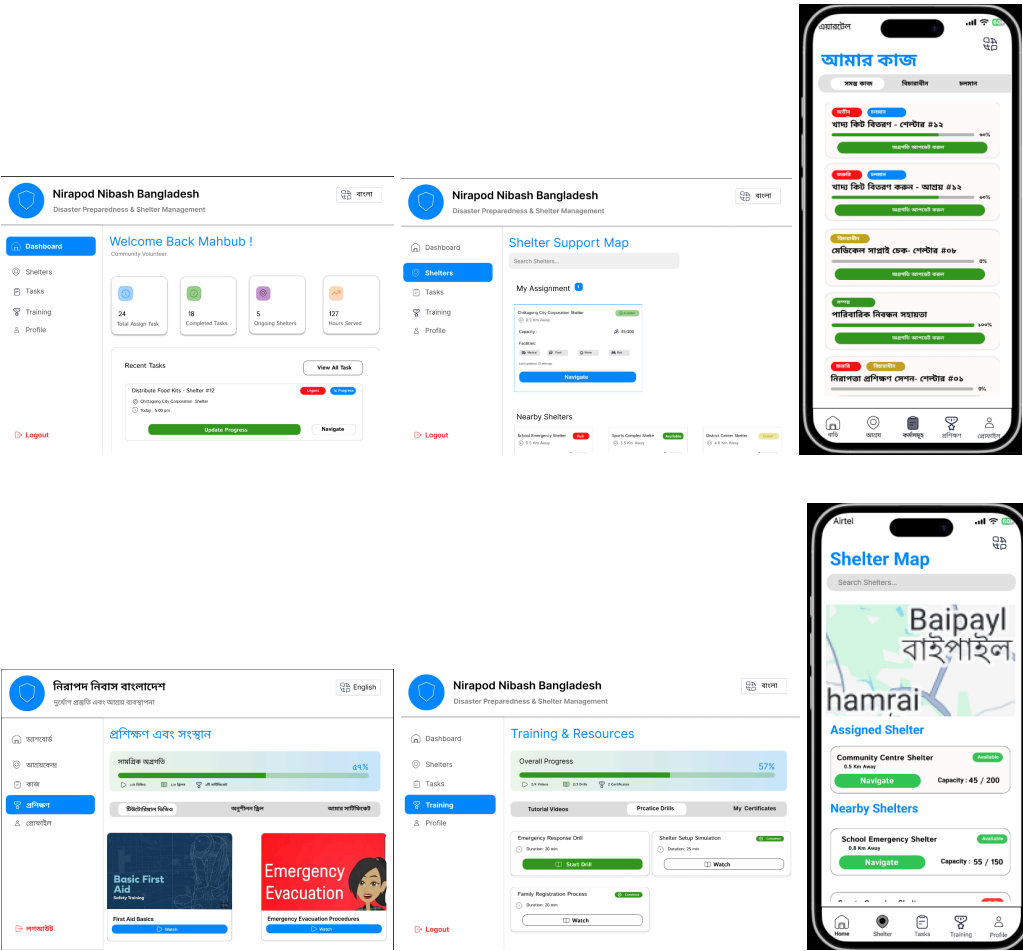


● Admin Interface





- Volunteer Interface



## 9. Conclusion

Nirapod Nibash Bangladesh provides a unified, data-driven system for managing disaster shelters. It bridges the coordination gap between citizens, volunteers, and authorities, enhancing national disaster resilience through accessible, real-time information.

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## 10. References

1. Government of Bangladesh, Disaster Management Bureau (2018–2023).
  2. IFRC Shelter Coordination Handbook (2021).
  3. BRAC & CARE Humanitarian Reports (2022).
  4. Dhaka University ICT in Disaster Relief Study (2022).
  5. UN OCHA Data Exchange Guidelines (2020).
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## 11. Appendices

### Appendix A – Glossary

- **NGO:** Non-Governmental Organization
- **GIS:** Geographic Information System
- **GPS:** Global Positioning System
- **SMS:** Short Message Service
- **API:** Application Programming Interface
- **DBMS:** Database Management System
- **DFD:** Data Flow Diagram
- **CRUD:** Create, Read, Update, Delete

